An Introductory and Comprehensive Guide to Using Schrödinger Maestro for Molecular Docking

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Specialty: Computational Chemistry, Artificial Intelligence and Machine Learning for Drug Discovery

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Preface

This guidebook is designed for beginners and early-career researchers in medicinal chemistry, pharmaceutical sciences, and computational drug discovery. It provides a step-by-step, end-to-end workflow for using Schrödinger Maestro in molecular docking for small-molecule drug discovery, from raw structural data to potential lead compounds. No prior experience with Schrödinger is assumed.

The guide emphasizes practical understanding, why each step matters, and best practices used in industry and academia.

Chapter 1: Overview of Schrödinger Maestro

1.1 What is Maestro?

Maestro is the graphical user interface (GUI) of the Schrödinger molecular modelling suite. It integrates multiple computational chemistry and molecular modelling tools into a unified workflow.

Maestro is commonly used for:

- Structure- and Ligand based drug design (SBDD) and (LBDD)
- Virtual screening
- Binding affinity estimation
- Binding Free Energy estimation.
- Pharmacophore modelling.
- Docking Validation.

1.2 Key Modules for Drug Discovery Researchers

PS: This guide focuses on Molecular docking and immediate rescoring techniques.

Module	Purpose
Protein Preparation Wizard	Clean and optimize protein structures
LigPrep	Prepare ligands (tautomers, protonation, stereochemistry)
Receptor Grid Generation	Prepare the grid for potential ligand binding
Glide	Molecular docking and virtual screening
Superimposition	Determine the RMSD for Docking Validation
Ligand Interactions	View 2D Interactions between ligands and target residues
Prime MM-GBSA	Binding free energy rescoring
Phase	Pharmacophore Modelling
QikProp	ADME and drug-likeness prediction
Enumeration	Bio-isosteric replacement

1.3 Downloading, Installing, and Opening Schrödinger Maestro. *Please see guide 2*

Chapter 2: Drug Discovery Workflow (Big Picture)

A typical Maestro-based drug discovery pipeline:

1. Define the biological problem
2. Prepare the protein structure
3. Prepare the ligand dataset
4. Identify or define the binding site
5. Screen large compound libraries or databases (High throughput Virtual Screening, Ligand based screening, etc)
6. Perform molecular docking at increasing precision levels (Standard and Extra)
7. Analyze and rank hits
8. Perform Binding free energy rescoring for top hits
9. Optimize leads
10. Predict ADME properties
11. Validate stability using molecular dynamics
12. Iterate and refine

This guide follows this exact sequence. Example workflows are shown below;

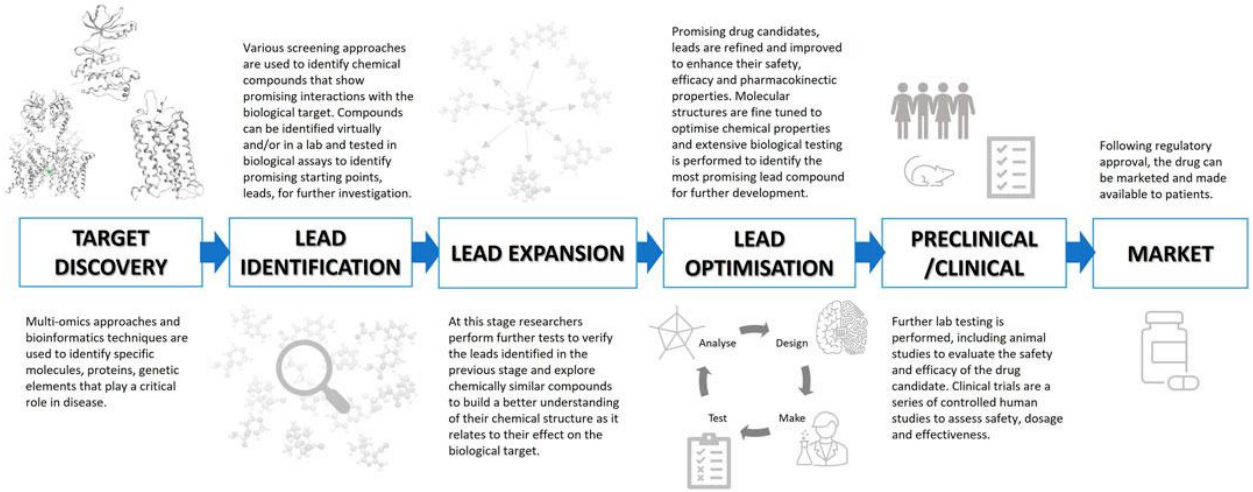


Fig 1A: Workflows in Computational Drug Discovery. *(Source: Southey MWY and Brunavs M (2023) Introduction to small molecule drug discovery and preclinical development.* Front. Drug Discov. *3:1314077. doi: 10.3389/fddsv.2023.1314077)*

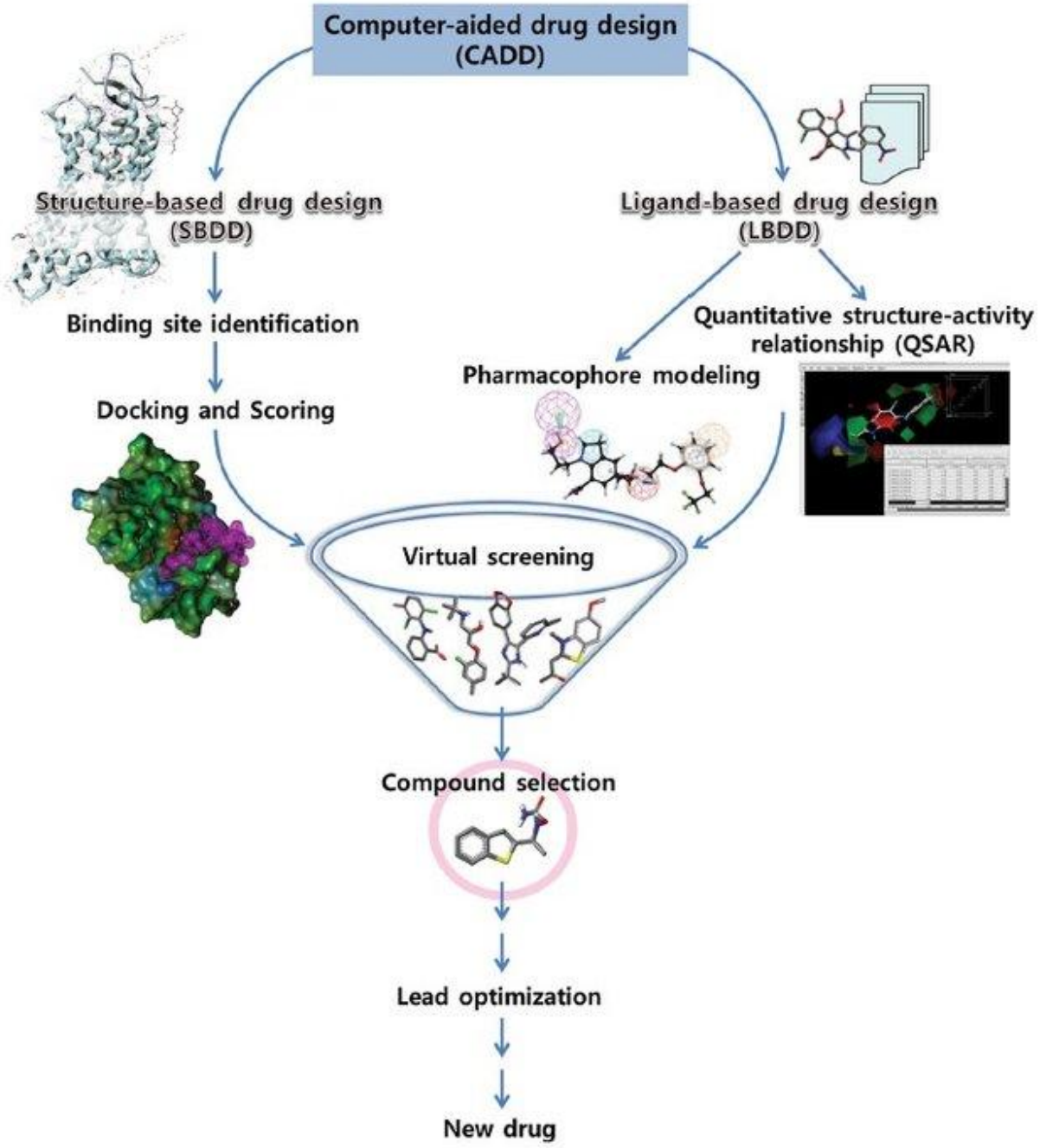


Fig 1B: Workflows in Computational Drug Discovery. (*Source: Prajapat P, Agarwal S, Talesara GL (2017) Significance of Computer Aided Drug Design and 3D QSAR in Modern Drug Discovery. J Med Org Chem Vol.1 No.1:1)*

Chapter 3: Protein Structure Preparation (CRITICAL STEP)

3.1 Why Protein Preparation is Essential

Raw PDB structures are not ready for docking. Common issues include:

- Missing hydrogens
- Incorrect bond orders
- Missing side chains
- Unresolved protonation states

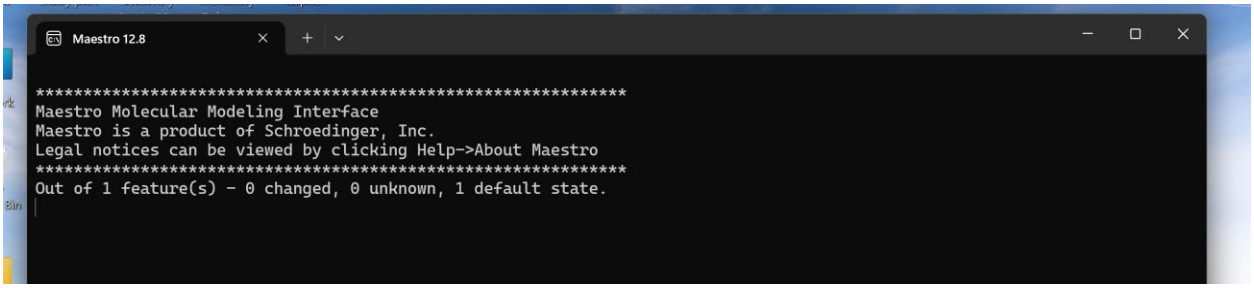
Improper preparation leads to false docking results.

3.2 Using the Protein Preparation Wizard

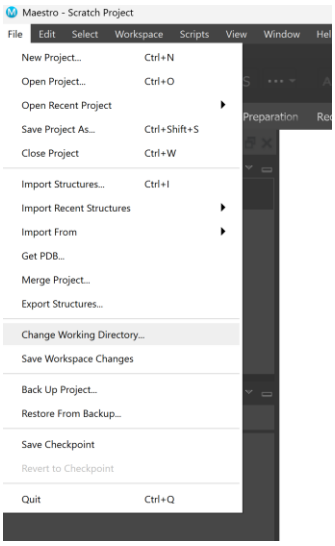
Tool: Protein Preparation Wizard

Step-by-Step

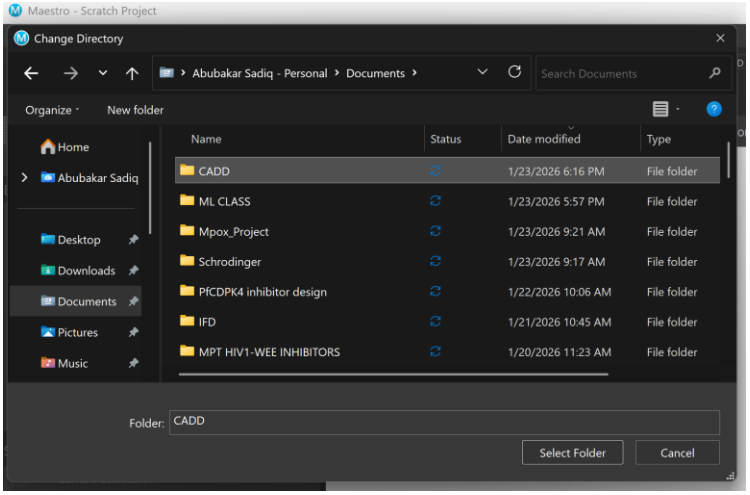
1. Download the protein structure (Example PDB ID: 1GS4) in its legacy PDB format from the Protein Data Bank via https://www.rcsb.org
2. Open your Computer's File Explorer. Create a new folder named 'CADD.' Transfer the downloaded PDB file from your downloads to the CADD folder.
3. Download, Install and Open the Maestro Schrodinger Suite. *(This guide is based on version 12.8)*, Allow Maestro to load and open.



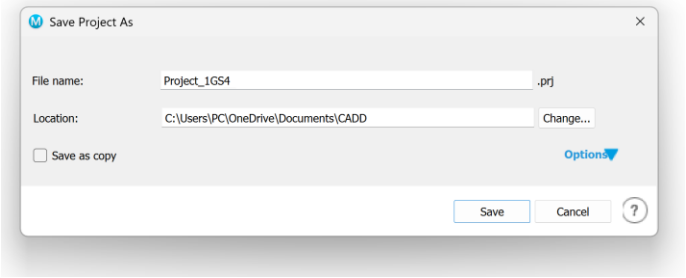
4. Navigate to File >>> Change Working Directory >>> CADD Folder



5. Select the CADD folder and Click on 'Select Folder'

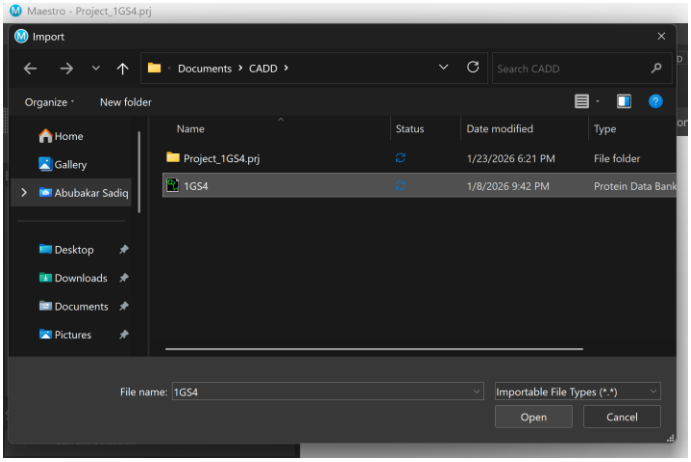


6. Navigate to File, click on 'Save Project As' and type 'Project_1GS4

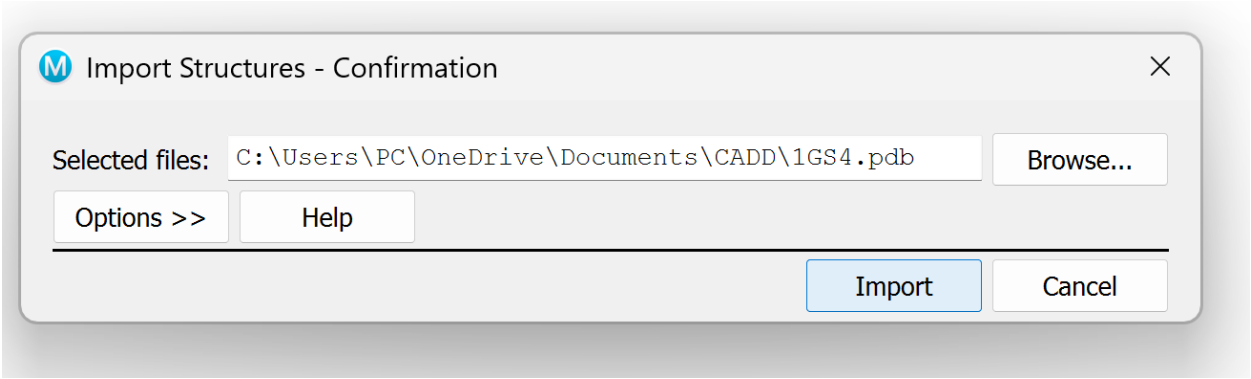


7. Navigate to File >>> Import Structures or Press Ctrl + I

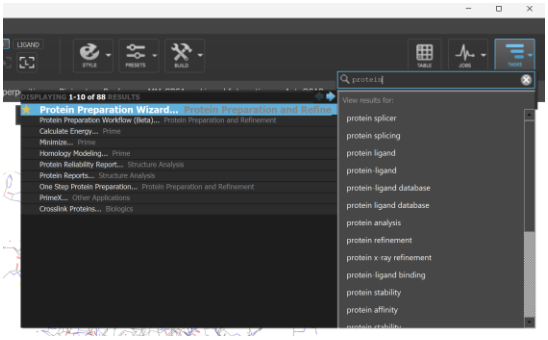
8. Select PDB file 1GS4, as shown below;



9. Click on ‘Open’.
10. Click on ‘Import’

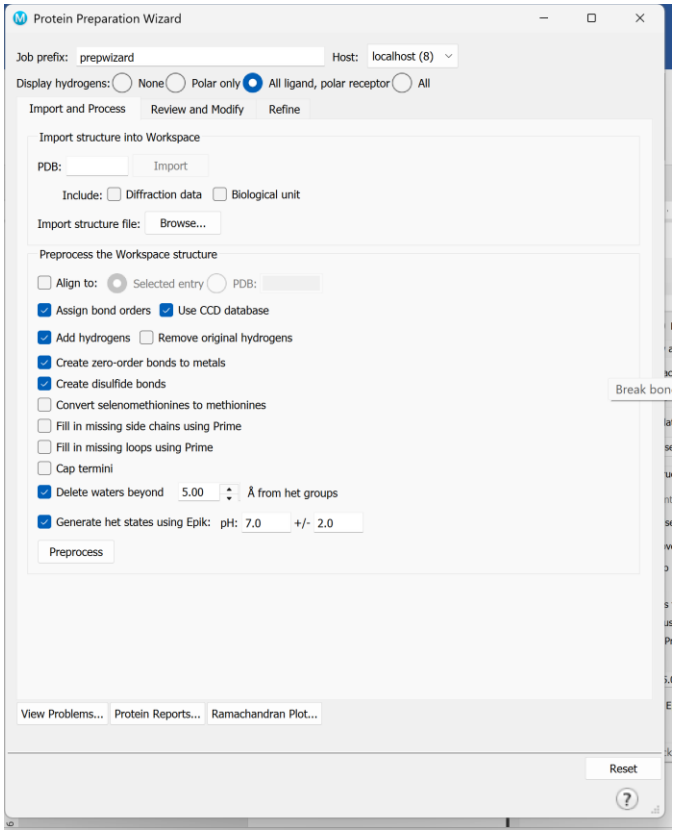


11. Navigate to the Task Bar in Maestro and search for ‘Protein’, select ‘Protein Preparation Wizard.’ as shown below

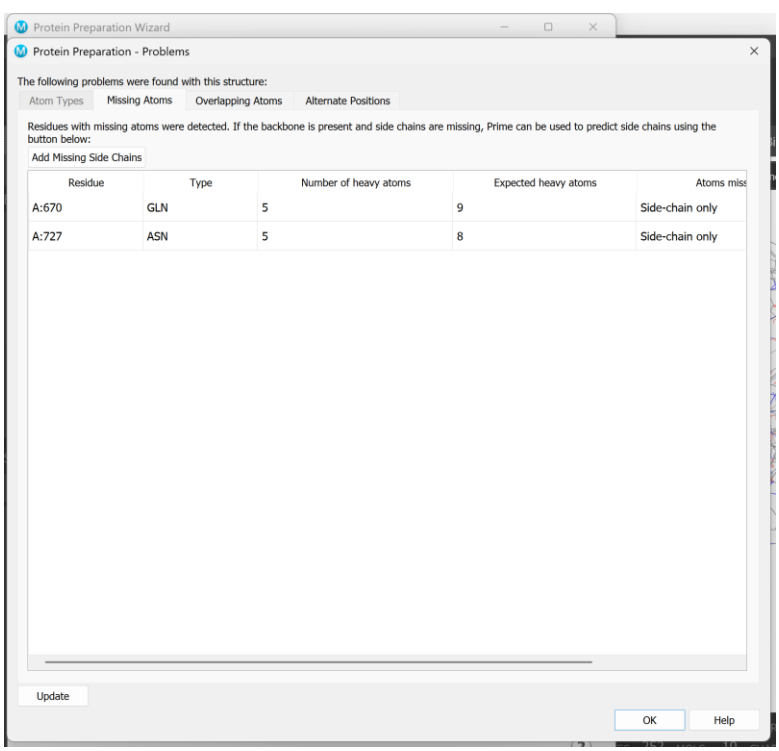


12. Preprocess the target using the following settings:
    - Assign bond orders
    - Add hydrogens
    - Create zero-order bonds to metals
    - Create disulfide bonds

- Delete waters beyond [5.00] Å from het groups
- Generate het states using Epik: pH 7.0 ± 2.0
- Click on Preprocess.

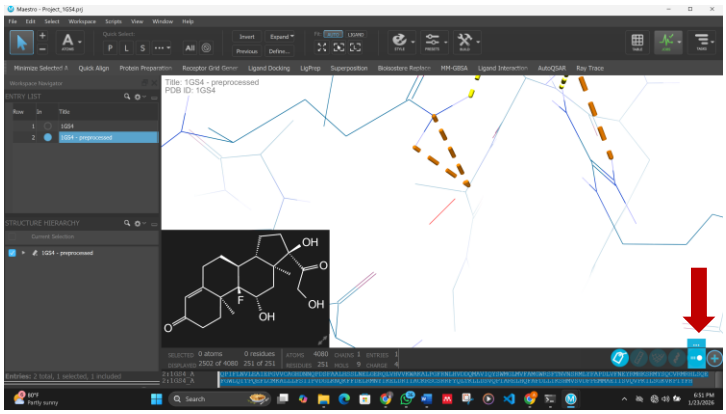


- If prompted, Click on ‘Update’ then ‘OK’. This automatically fixes any structural problems in the target as identified by Scrodinger.

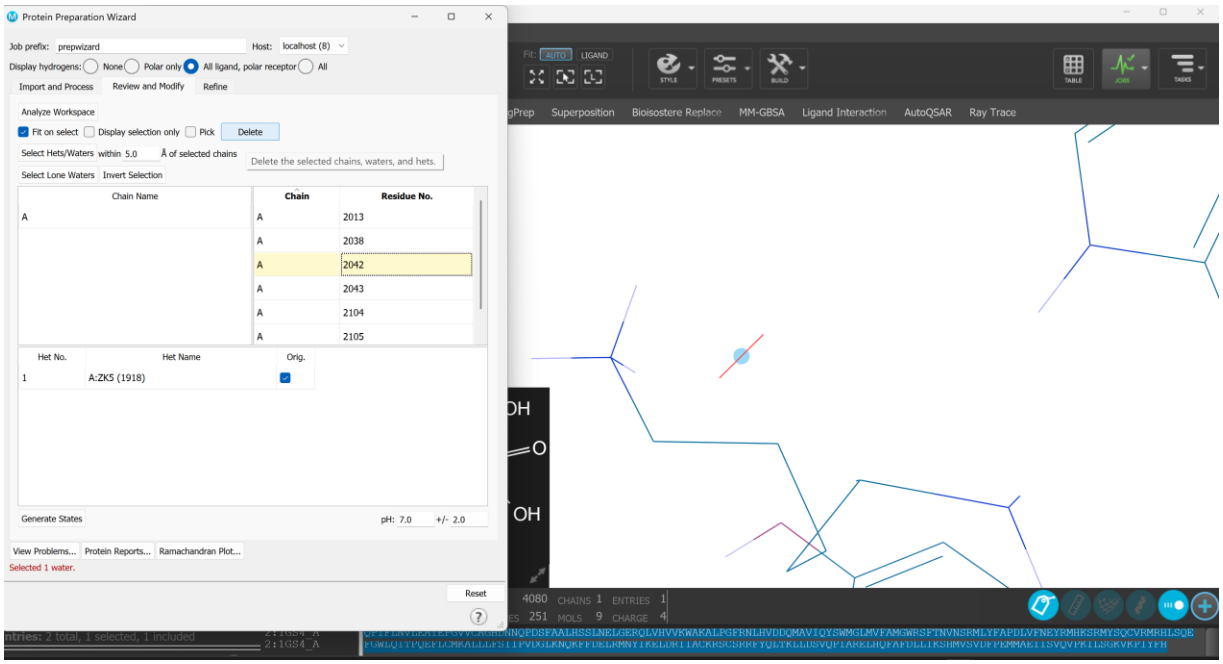


13. Review and Modify the target **based on the information in the literature of the target.** This literature contains information on conserved water(s), heteroatoms, binding site residues and pockets, which are vital for the correct functioning of the protein. Review and modify using the following settings:

- Click the 'Interactions Toggle' to turn it on.

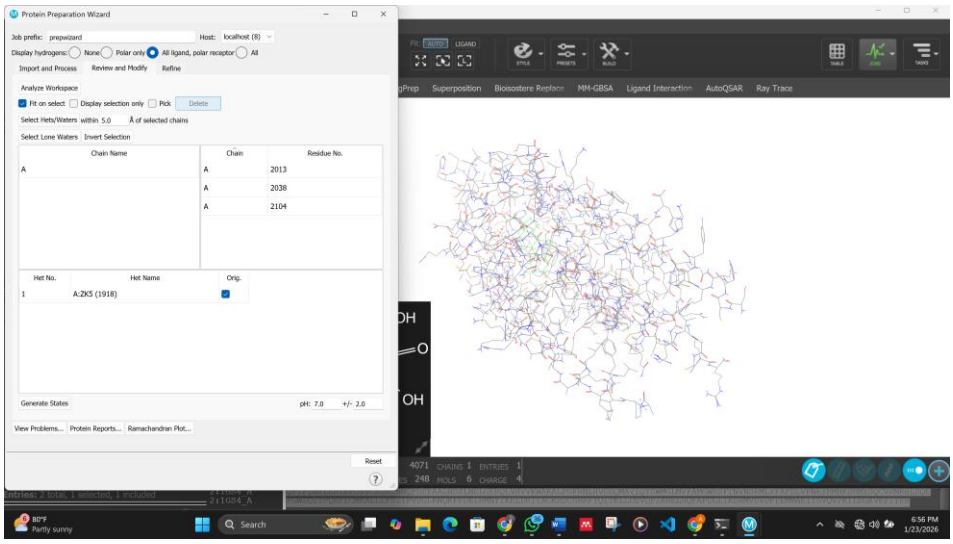


- Delete **PO4** (Phosphate) heteroatom. *(it is not necessary for the binding activity of target, based on information gotten from the literature)*
- Delete all water molecules that do not form an interaction with the ligand or target, as shown below;



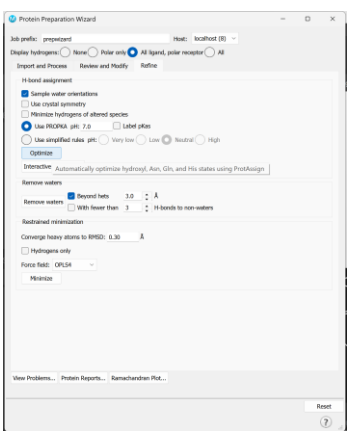
- Where duplicate chains of same functionality exist, delete the duplicate chains and have just one chain. (Not applicable for this protein.)

- If done correctly, your screen should look like this at the end of this stage;



14. To automatically optimize hydrogen bond network, hydroxyl, and amino acid states, use the following settings;

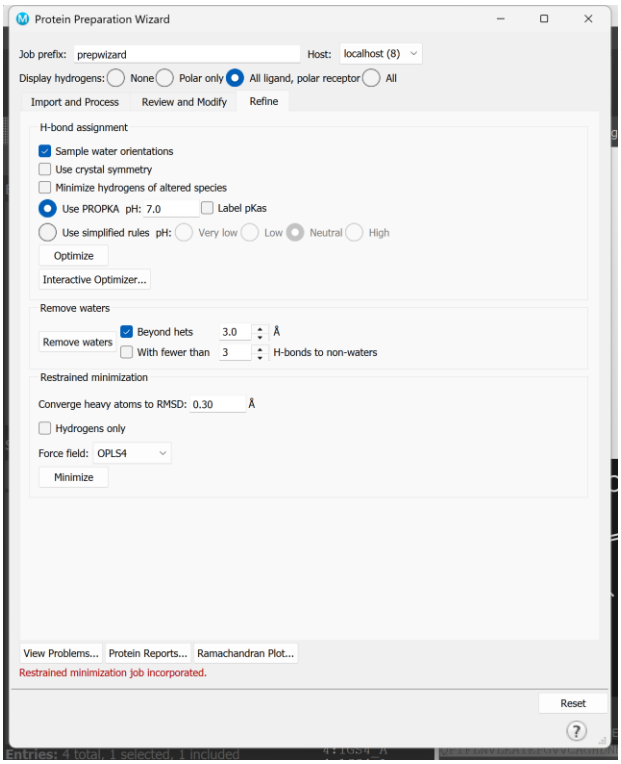
    - Display hydrogens: All ligand, polar receptor
    - Sample water orientations
    - Use PROPKA pH 7.0
    - Optimize.



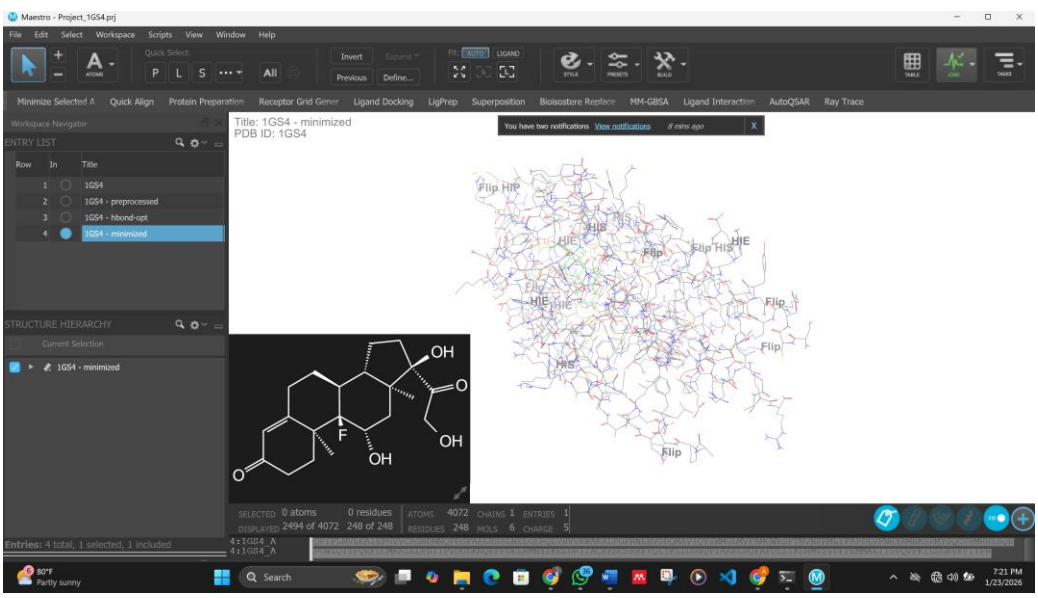
15. Allow the optimization task to run and incorporate.
16. Run a restrained minimization job, with heavy atoms restrained (or frozen) and hydrogens unrestrained using the following settings;

    - Converge Heavy atoms to RMSD: 3.00 Å
    - Force field: OPLS4
    - Minimize

17. Allow the minimization task to run and incorporate.



18. On successful completion, your entry table will look like the image below;



3.3 Beginner Tips

- Always inspect the structure visually
- Retain crystallographic waters involved in ligand binding
- Verify cofactors and metal ions

Chapter 4: Ligand Preparation

4.1 Why Ligand Preparation Matters

Ligands may exist in multiple:

- Protonation states
- Tautomers
- Stereochemical forms

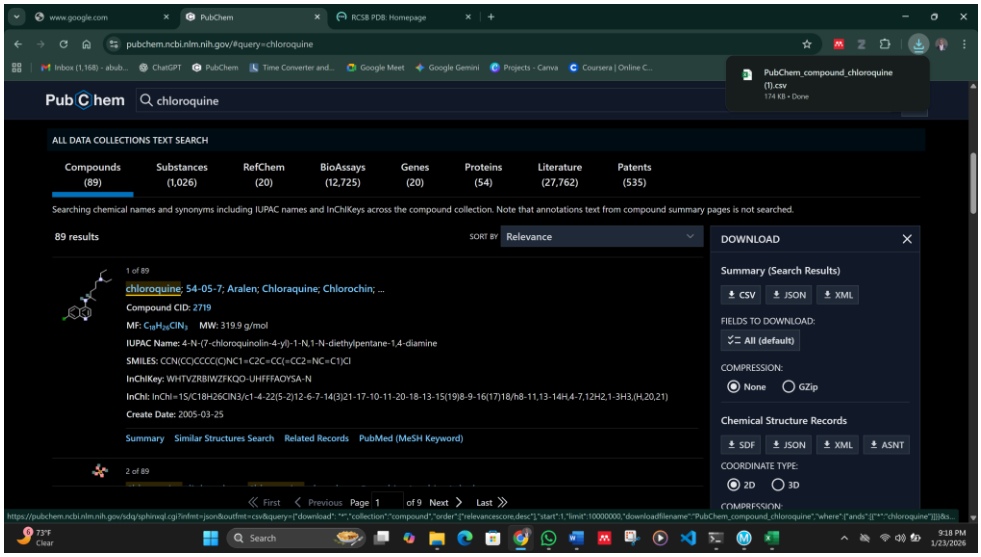
Docking only one form can miss the true bioactive conformation.

4.2 LigPrep Workflow

Tool: LigPrep

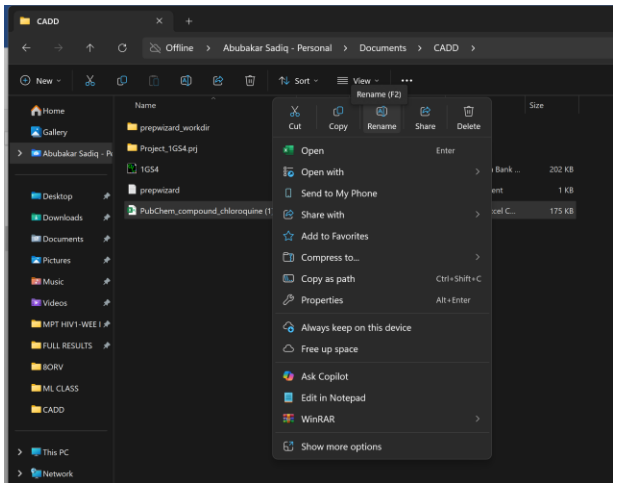
Steps:

1. Collate ligands to be investigated. Below is an example, assuming we want to repurpose Chloroquine compounds for the treatment of prostate cancer;
    A. Log on to the PubChem database using (https://pubchem.ncbi.nlm.nih.gov/).
    B. Search for ‘Chloroquine’
    C. Scroll down and click on ‘Download’
    D. Click on Download as CSV



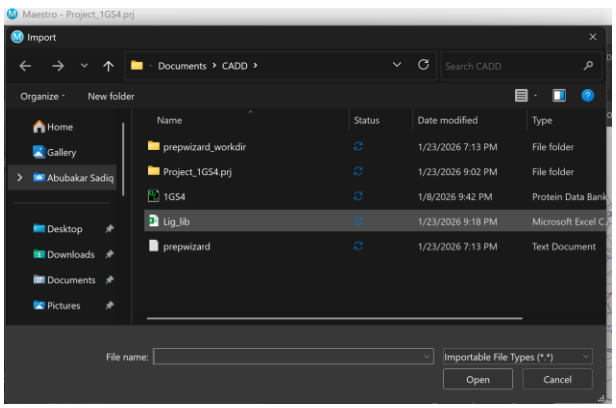
E. Transfer the downloaded file to the CADD folder on your computer.

F. Rename the downloaded PubChem file to ‘Lig_lib’



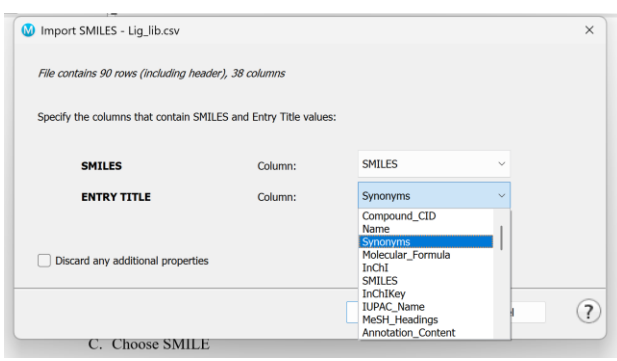
2. Import ligands (SMILES, SDF)

A. Navigate to Files >>> Import Structures >>>Select Lig_Library



B. Click on ‘Open’ >>> Import.

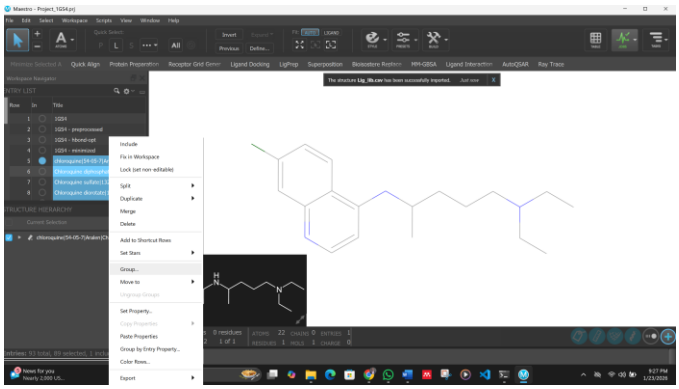
C. Choose SMILES Column as ‘SMILES’ and ENTRY TITLE Column as ‘Synonyms’



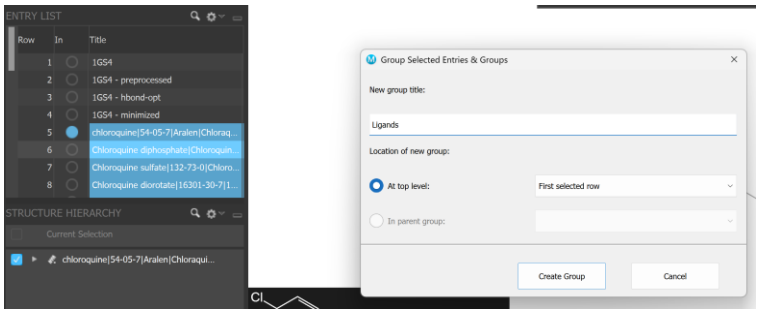
3. Organize your entry table;

A. Right click on the highlighted newly imported ligands on your entry list.

B. Click on ‘Group’



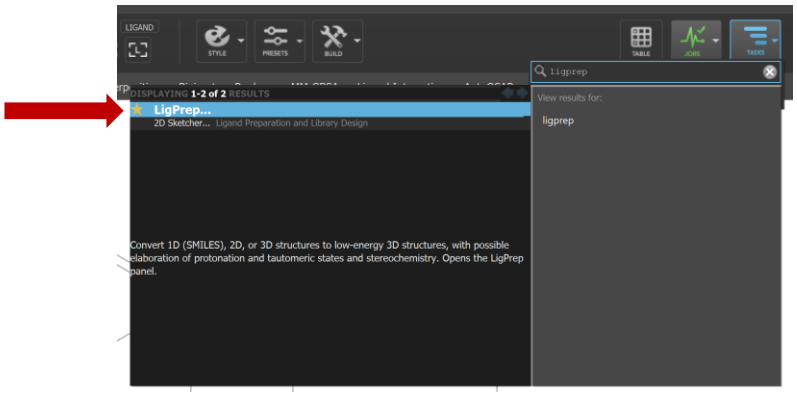
C. Title your new entry list as ‘Ligands.’ Click ‘Create Group.’



D. **NB:** You can expand or collapse the entry list by using the arrow shaped toggle.

4. Prepare Ligands using the Maestro LigPrep module

A. Navigate to tasks >>> Search for ‘ligprep’ >>> Click on it to open

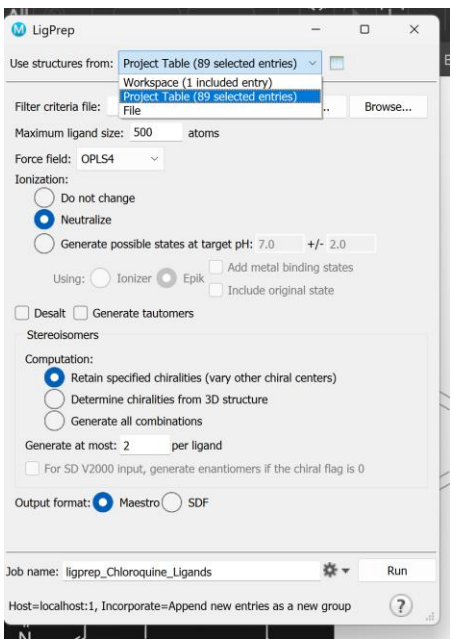


NB: You can fix tools to your Quick access menu by clicking on the Star for the tool.

B. Ensure the ligands are all selected on the entry list

C. Prepare the ligands using the following settings;

- Use Structures from: ‘Project Table’
- Maximum ligand size: 500
- Force field: OPLS4
- ***(NB the following parameters can change based on your research hypothesis)***
    1. Ionization: Neutralize
    2. Uncheck ‘Desalt’ and ‘Generate Tautomers’
    3. Stereoisomers: Computation- retain specified chiralities. Generate at most [2] ligands. *(Higher number of stereoisomers can be explored. >5 stereoisomers per ligand is recommended.)*
- Output format: Maestro
- Job name: ligprep_Chloroquine_Ligands
- Click on run. Allow task to be executed and incorporated.



4.3 Best Practices

- Limit excessive enumeration to reduce computational cost
- Keep biologically relevant protonation states

Chapter 5: Binding Site Identification

5.1 Known Binding Site

- Use cognate ligand
- Define grid around ligand centroid

5.2 Unknown Binding Site

- Define grid around Selected residues centroid
- Define grid from previously known/determined x,y,z coordinates.

Tip: For beginners usage of a target with cognate ligand is recommended to ease binding site identification.

Chapter 6: Molecular Docking with Glide

6.1 Grid Generation

Tool: Receptor Grid Generation

Key settings:

- Center on ligand or key residues
- Inner box: binding region
- Outer box: ligand exploration space

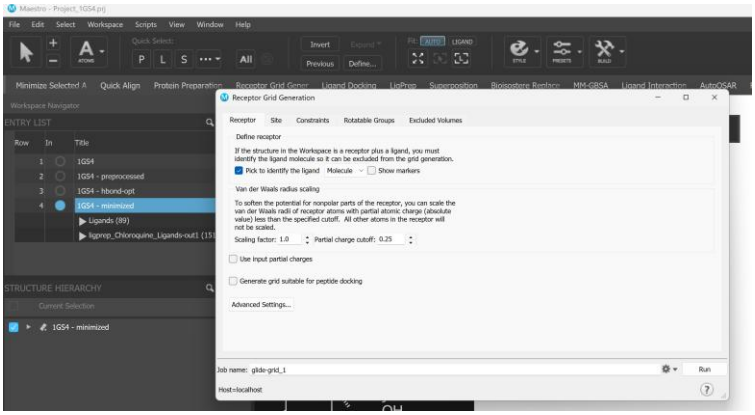
Practical:

A. Include the minimized protein-ligand complex in the workspace

B. Navigate to the Tasks Menu.

C. Search for ‘Receptor grid generation’

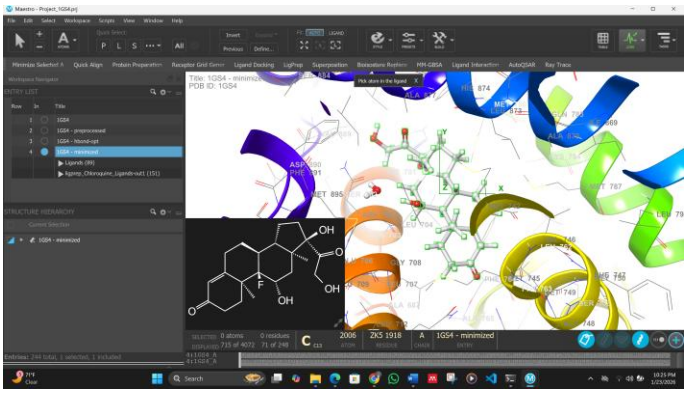
D. Navigate to the ‘Receptor (Define Receptor)’ box, ensure that ‘Pick to Identify Ligand [Molecule]’ is checked.



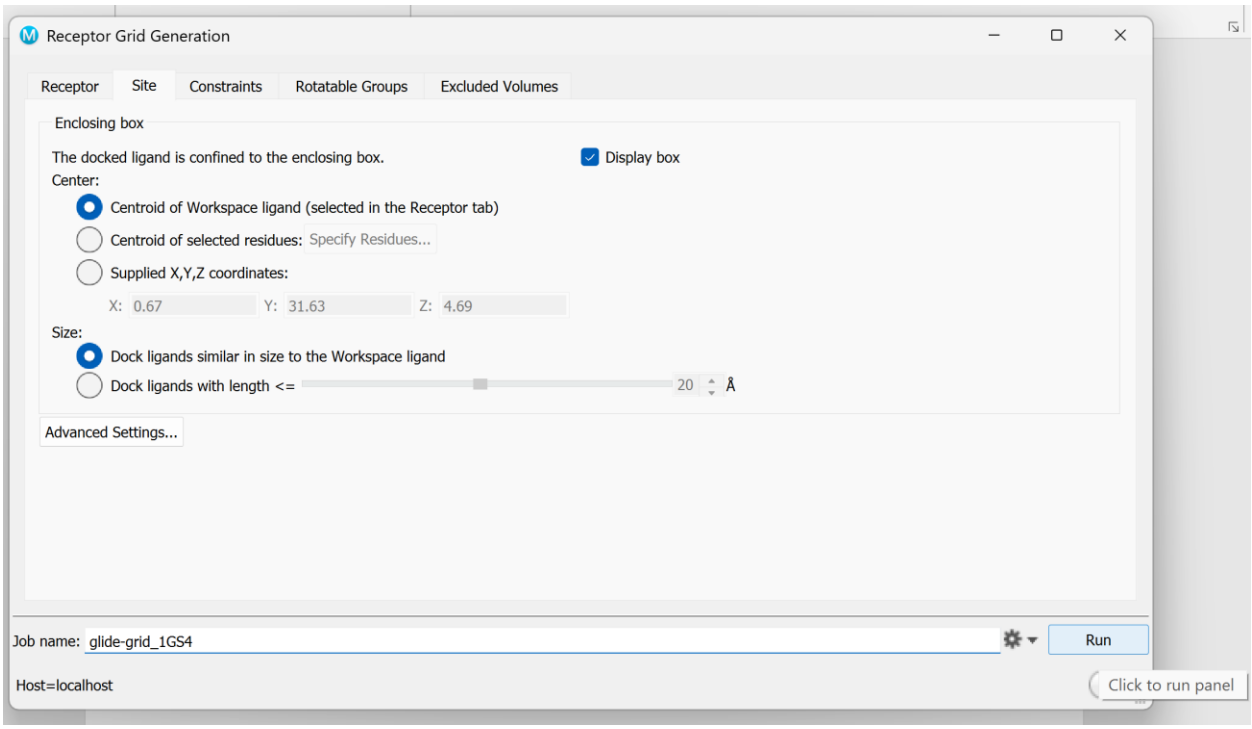
E. Navigate to Enclosing box and ensure that ‘Centroid of Workspace Ligand’ is selected.

F. Navigate to the Maestro workspace and click on any atom of the ligand.

Tip: Press ‘L’ or the [L] shortcut on the tools bar to zoom and fit the ligand to the screen.



G. Return the ‘Receptor Grid Generation’ module, note the new grid x,y,z coordinates, rename the grid file to the protein’s PDB ID ‘1GS4’ and ‘Run’.



H. Allow the job to run and incorporate.

6.2 Docking Modes Explained

Mode	Recommended Use Case	Speed	Accuracy
HTVS	Large libraries (>100k)	Fastest	Lowest
SP	Moderate screening	Moderate	Moderate
XP	High-accuracy refinement	Slowest	Highest

6.3 Recommended Docking Workflow

1. HTVS screening >>> Select top 10–20%

   >>> SP docking >>> Select top 10–20%

   >>> XP docking for final hits

6.4 Interpreting Docking Scores

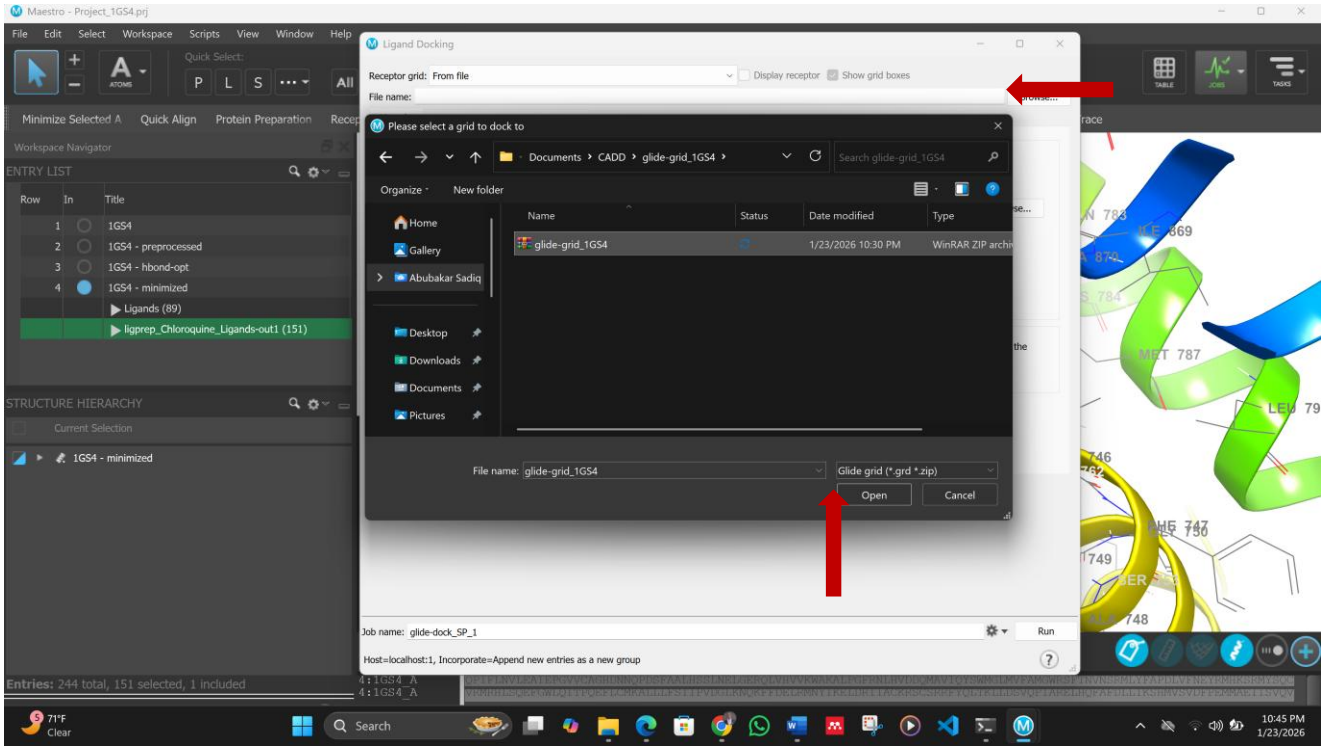
- More negative GlideScore = stronger predicted binding

To execute:

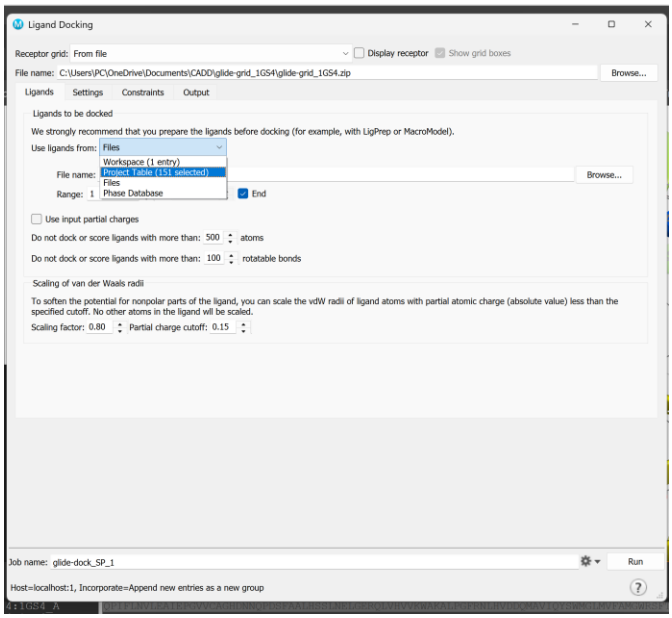
A. Select the prepared ligands on the entry list.

B. Navigate to Tasks >>> Ligand Docking

C. **For Receptor Grid:** File name >>>Browse >>>Sselect ‘1GS4’ >>> Open



D. For Ligands: Use Ligands from the ‘Project Table’



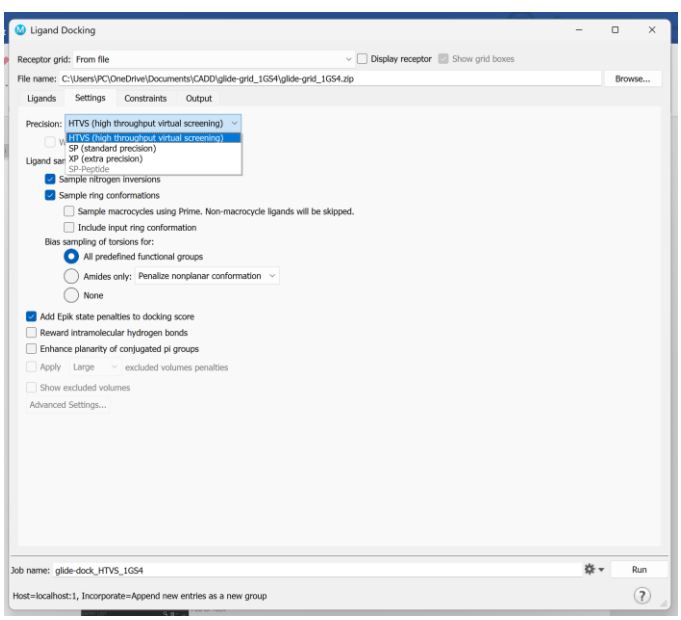
E. **For Settings;**

- Precision**:** Select ‘HTVS (high throughput virtual screening)
- Ligand Sampling: Flexible
- Check Sample nitrogen inversions
- Check sample ring conformations
- Bias sampling of torsions for ‘All predefined functional groups.
- Check ‘Add Epik state penalties to docking score.

F. Rename the Job name as; glide-dock_HTVS_1GS4

G. Allow the job to run and incorporate.

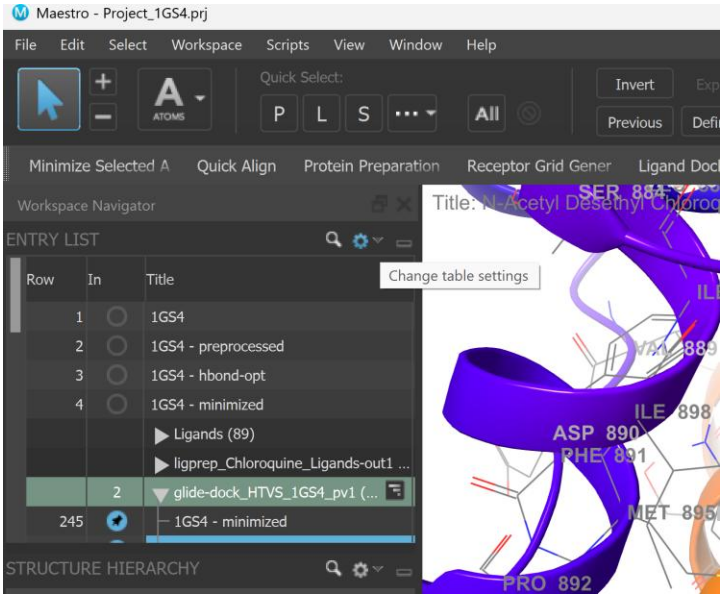
H. Repeat docking using higher precisions of SP and XP for the top 10-20% compounds with the highest docking scores.



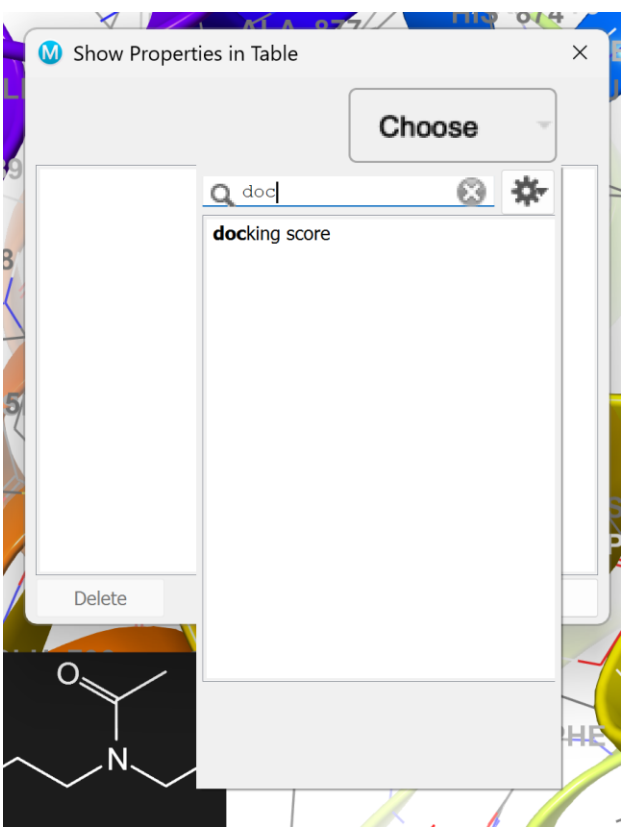
Chapter 7: Post-Docking Analysis

7.1 Viewing your docking score

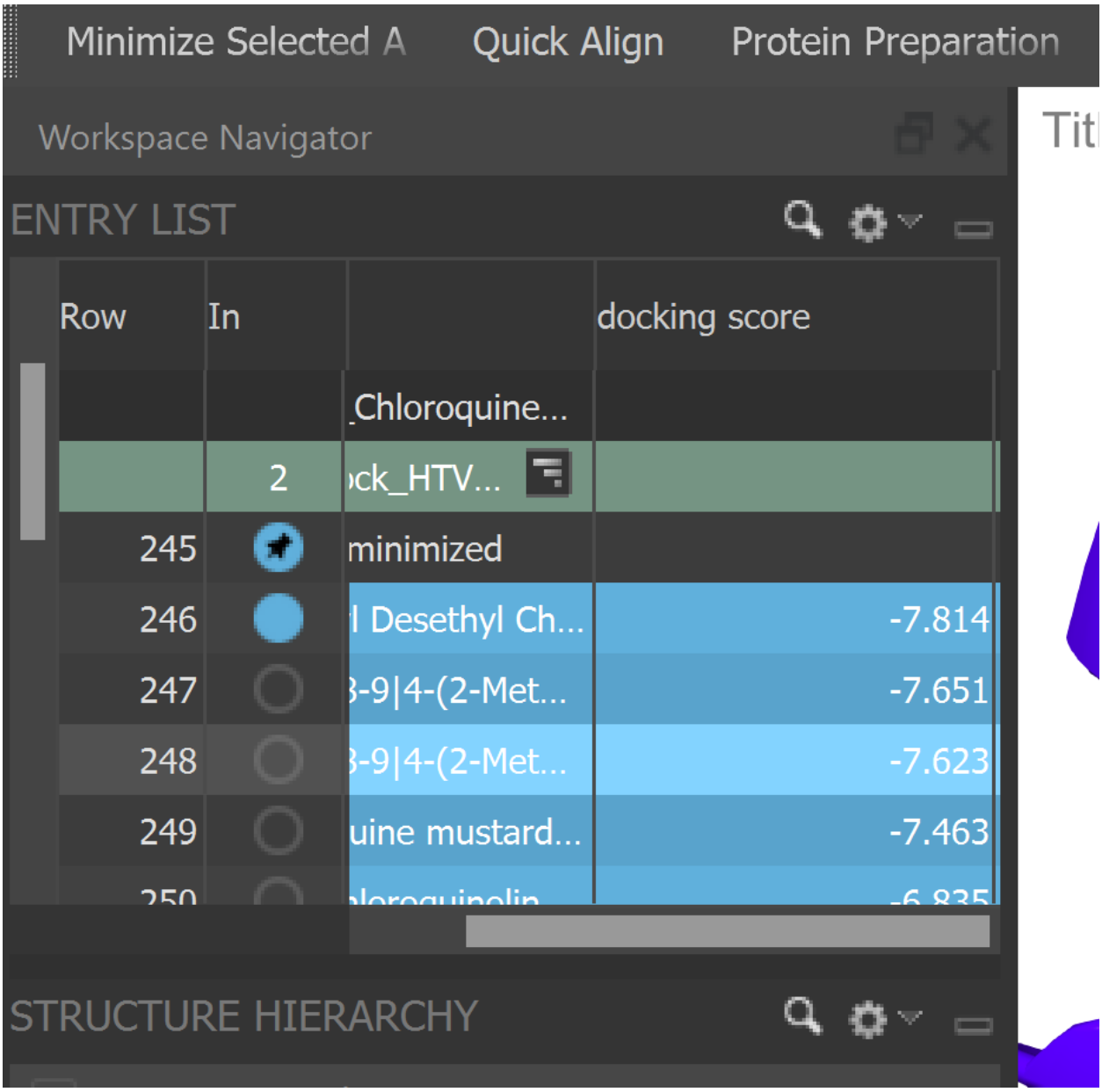
- Click on the entry list table settings.



- Select ‘Show Property’. From the ‘Choose’ menu search for ‘docking Score’ and click ‘OK’

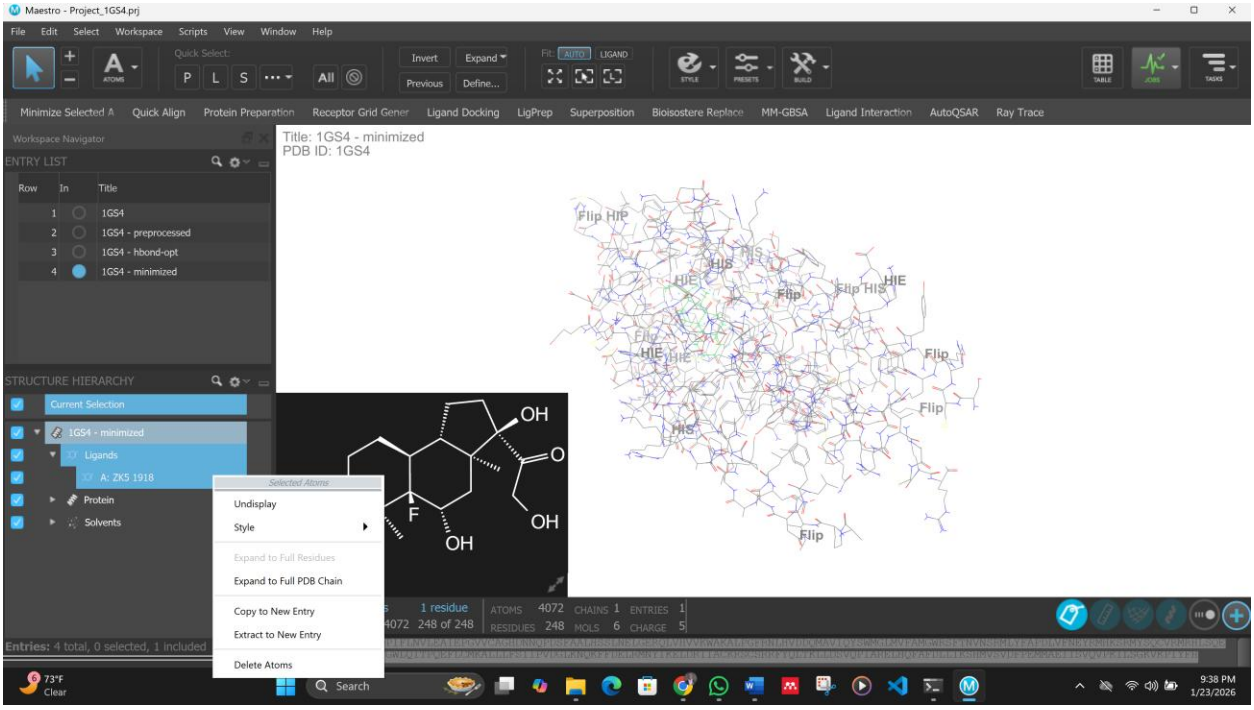


- Your entry table now displays docking score, after the title of each docked entry.

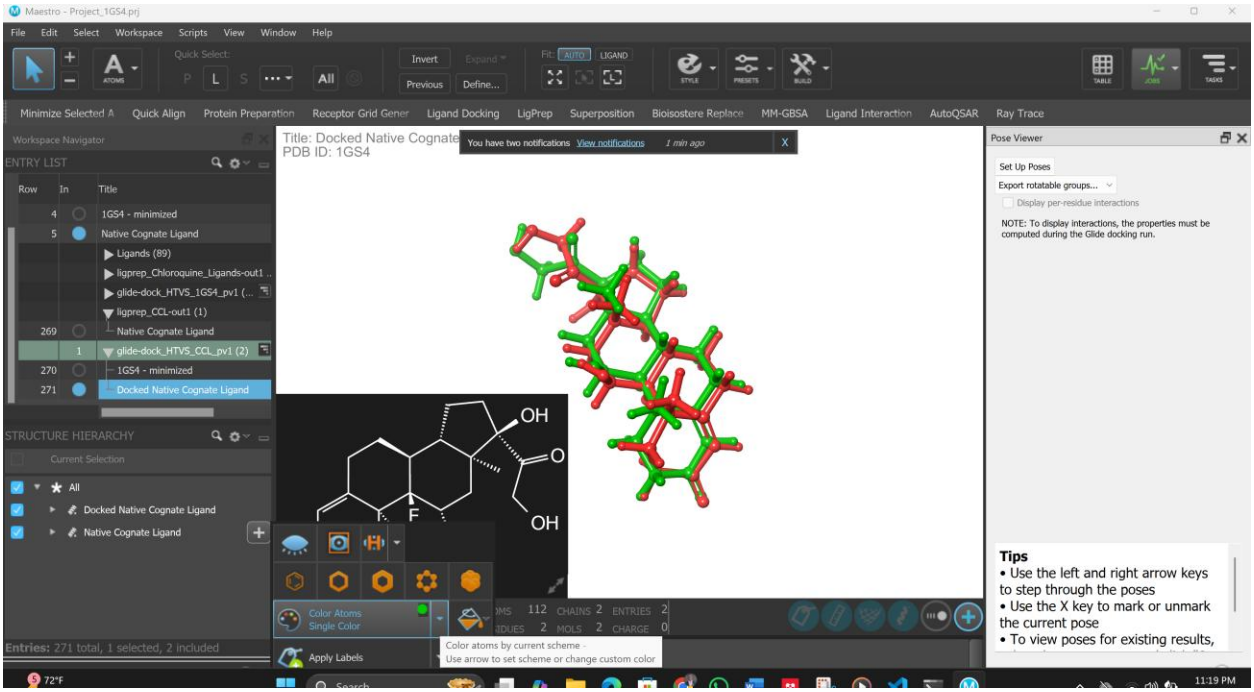


7.2 Validating your docking protocol

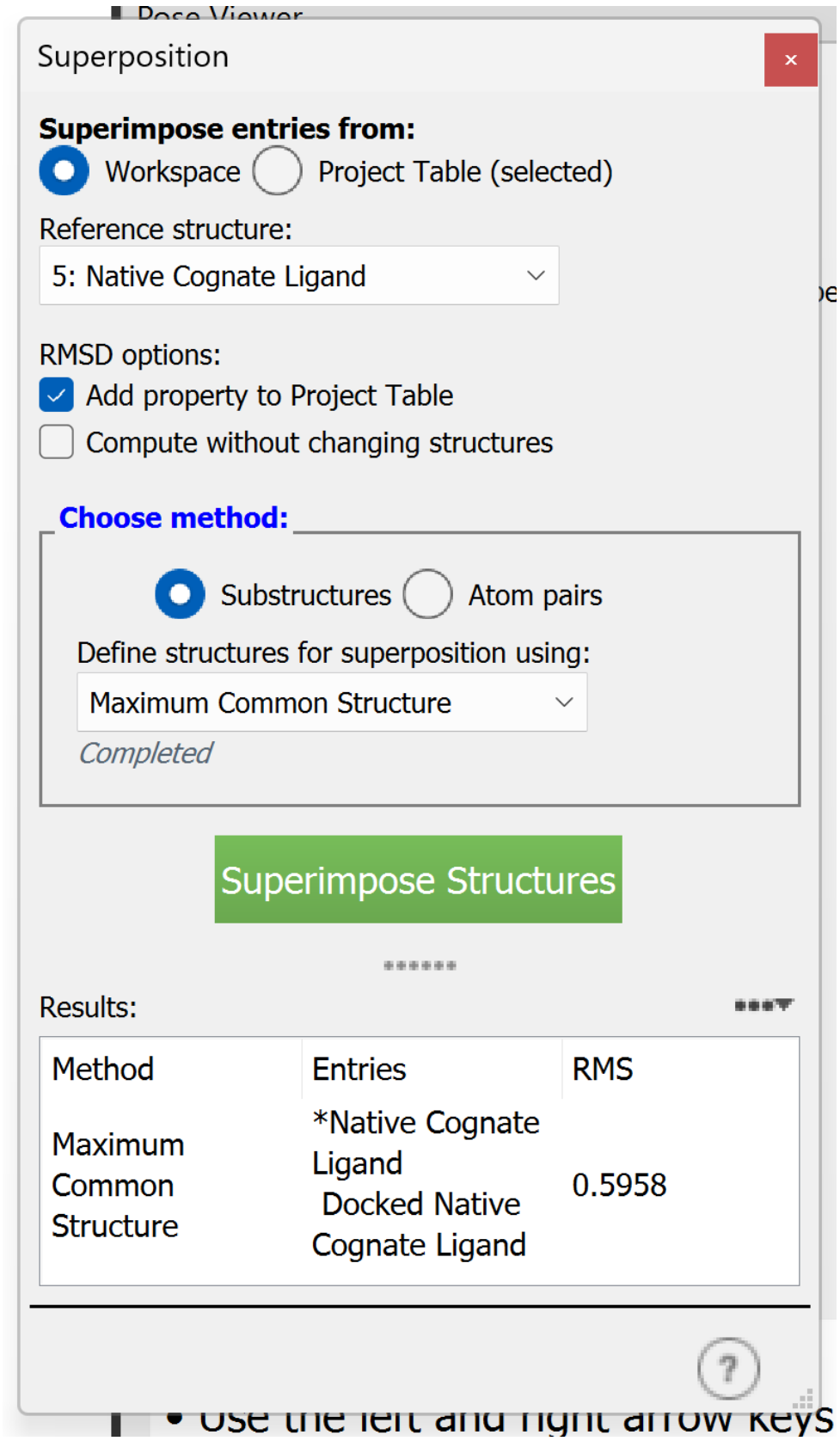
- NB: It is recommended to validate your docking protocol before docking test ligands, to verify if you have optimally prepared the target and if your planned setup will produce reliable results. This aspect of the tutorial has been placed at this point for simplicity.
- To validate the docking protocol, follow the instructions below;
    - Include the minimized protein-ligand complex into your workspace.
    - Expand the minimized complex entry on the workspace until the 'ZK5 1918' Cognate ligand entry is on display, as shown below;



- Right click on the cognate ligand and click on ‘Copy to New Entry.’
- Rename ‘Structure**’ to ‘Cognate Ligand’.
- Prepare the cognate ligand as described above
- Dock the prepared cognate ligand using the HTVS, SP and XP docking precisions, as described above.
- Include the native original cognate ligand and the redocked cognate ligand to the workspace.



- Navigate to Tasks >>> Superimposition
- Superimpose entries from ‘[Workspace]’, ‘RMSD: [Add property to Project Table]’, ‘Choose method: [Substructures]’ , ‘Define structures for superimposition using [Maximum Common Structure]’
- Click on ‘Superimpose Structures

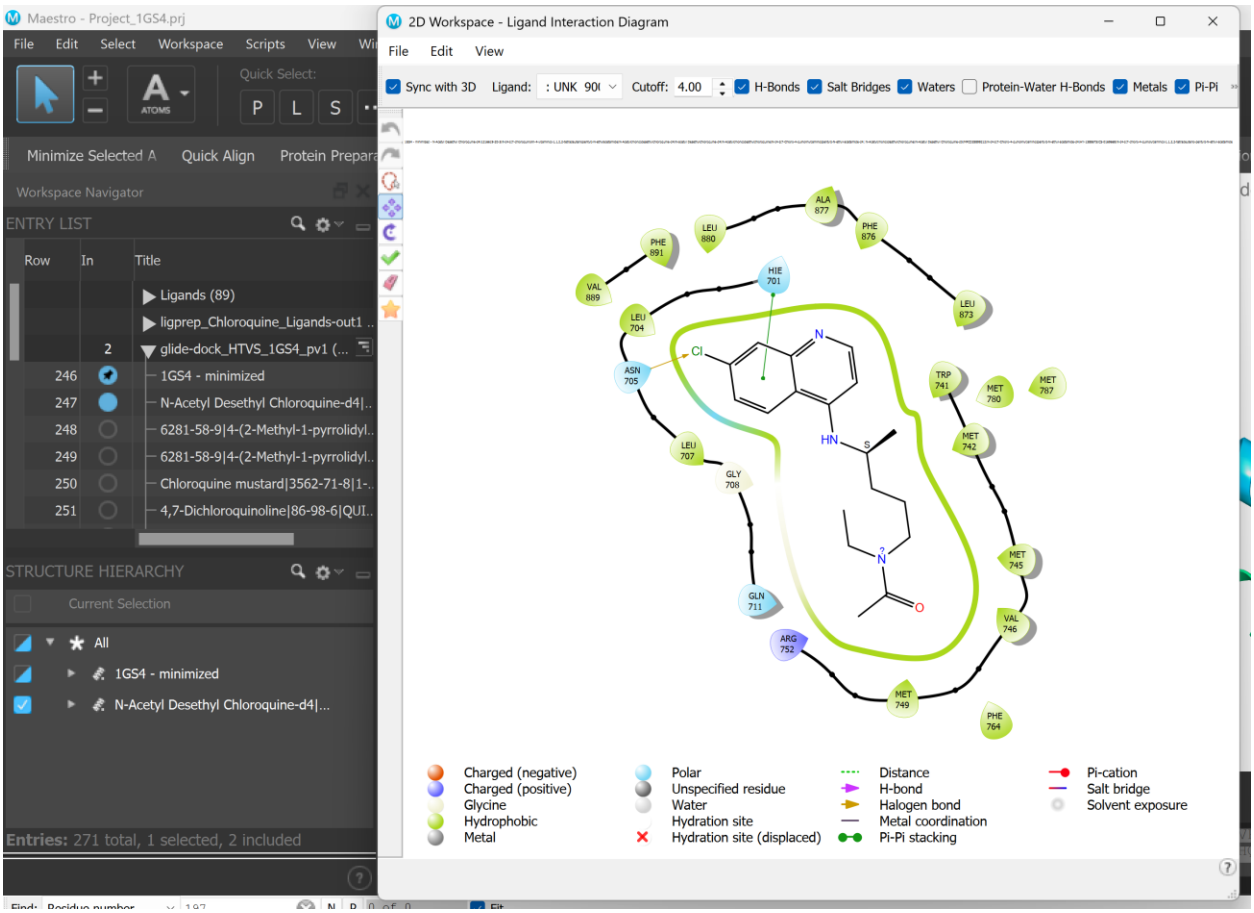


- Use the left and right arrow keys

7.3 Interaction Diagrams

Use Maestro interaction diagrams to communicate binding modes clearly.

- Include a docked ligand along with its target
- Navigate to Tasks >>> Ligand Interactions



Chapter 8: Reporting and Reproducibility

Include:

- Software version
- Parameters used
- Clear figures and tables

Chapter 9: Common Beginner Mistakes

- Skipping protein preparation
- Trusting docking scores blindly
- Ignoring protonation states
- Over-interpreting weak predictions

Selected References

1. Friesner, R. A. et al. *Glide: A New Approach for Rapid, Accurate Docking and Scoring.* J. Med. Chem., 2004.
2. Halgren, T. A. et al. *Glide XP Docking and Scoring.* J. Med. Chem., 2006.
3. Sastry, G. M. et al. *Protein and Ligand Preparation.* J. Comput.-Aided Mol. Des., 2013.
4. Jacobson, M. P. et al. *A Hierarchical Approach to All-Atom Protein Loop Prediction.* Proteins, 2004.
5. Schrödinger Release Manuals, Schrödinger LLC.
6. Jorgensen, W. L., Tirado-Rives, J. *Potential Energy Functions for OPLS Force Fields.* JACS, 1988.
7. Abel, R. et al. *FEP+: A Modern Free-Energy Calculation Tool.* Acc. Chem. Res., 2017.